MarketSight

Stock Market Prediction System

Vision Document

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1. Introduction

1.1 Purpose

This document describes the vision for MarketSight, a web-based stock market prediction platform that uses machine learning to forecast stock prices. It captures the stakeholder needs, high-level features, and constraints that shape the system.

1.2 Scope

MarketSight is a capstone project for COSC 471 at Okanagan College. The system provides end-to-end stock price prediction with confidence scoring, interactive data visualization, automated data pipelines, and production deployment. The project spans three construction sprints (Sprints 3-5, January-March 2026).

1.3 References

- IBM Rational Unified Process (RUP) Vision template
- Agile Unified Process (AUP) guidelines
- COSC 471 course deliverable requirements

2. Positioning

2.1 Business Opportunity

The retail investment technology market continues to grow as individual investors seek data-driven tools. Most ML-based prediction platforms are either proprietary (expensive) or academic (no production deployment). MarketSight fills the gap as an educational, open, and deployable platform that demonstrates a complete ML prediction pipeline from data ingestion to user interface.

2.2 Problem Statement

Element	Description
The problem of	Lack of accessible, transparent stock prediction tools
Affects	Retail investors, finance students, and data analysts
The impact of which is	Reliance on gut feeling, expensive subscriptions, or opaque prediction tools
A successful solution would	Provide free ML-powered predictions with explainable confidence scores and interactive visualizations

2.3 Product Position Statement

Element	Description
For	Retail investors, finance students, and data analysts
Who	Need data-driven stock price forecasts with reliability indicators
MarketSight	Is a web-based stock prediction platform
That	Uses XGBoost ML models trained on 53 technical indicators to predict prices with confidence scores
Unlike	Expensive proprietary platforms or black-box prediction tools
Our product	Is transparent, educational, and deployable with automated data pipelines

3. Stakeholder and User Descriptions

3.1 Stakeholder Summary

Stakeholder	Role	Responsibilities
Professor (Client)	Course instructor, project sponsor	Evaluates deliverables, signs acceptance tests
Assistant Instructor	Teaching assistant	Reviews submissions, provides feedback
Zane Tessmer	Scrum Master	Sprint ceremonies, documentation, coordination
Harsh Kumar	Product Owner / Lead Developer	Backlog prioritization, architecture, ML pipeline
Dante Bertolutti	Developer	Confidence scoring, predictions UI
Parag Jindal	Developer	Frontend routing, API client
Kaval S	Developer	User settings, dashboard optimization
Guntash	Developer	Dashboard cards, UI polish

3.2 User Summary

User Type	Description	Responsibilities
Retail Investor	Individual seeking price forecasts	Browse stocks, generate predictions, interpret confidence
Day Trader	Active trader needing quick forecasts	Batch predict, compare returns, view charts
Finance Student	Learner studying quantitative analysis	Explore data, understand indicators, study model behavior
Data Analyst	Technical user exploring market data	Use Streamlit for ad-hoc queries, dataset management
Administrator	System admin managing user accounts	Add/edit/delete users, manage permissions

3.3 User Environment

- Users access MarketSight through a modern web browser (Chrome, Firefox, Edge, Safari)
- The React dashboard runs at the deployed URL or `http://localhost:5173` in development
- The Streamlit data explorer runs separately for analyst workflows

- Administrators need superuser credentials to access the admin panel
- No software installation is required for end users

3.4 Key Stakeholder Needs

Need	Priority	Current Solution	Proposed Solution
Stock price predictions	HIGH	Manual analysis or paid tools	Automated ML predictions via XGBoost
Prediction reliability	HIGH	None (blind trust)	Confidence scoring (0-100%) with color coding
Data visualization	HIGH	Third-party charting tools	Built-in candlestick/line charts with overlays
Batch predictions	MEDIUM	Run predictions one at a time	"Predict All" feature for all tracked stocks
Automated data freshness	MEDIUM	Manual data downloads	Airflow DAGs for daily seeding and weekly retraining
Data exploration	MEDIUM	SQL queries	Streamlit frontend for visual exploration

4. Product Overview

4.1 Product Perspective

MarketSight is a standalone web application. It does not integrate with third-party trading platforms or broker APIs. Market data is sourced from the Financial Modeling Prep (FMP) API. The system consists of:

- **React 19 Frontend** - TypeScript, Vite, Tailwind CSS, TanStack Router
- **FastAPI Backend** - Python 3.12, SQLAlchemy 2.0, Pydantic v2
- **PostgreSQL 16 Database** - Market data, user accounts
- **XGBoost ML Model** - Native .ubj format, Optuna-optimized, 53 features
- **Apache Airflow** - Data seeding and model retraining DAGs
- **Nginx** - Reverse proxy with SSL termination
- **Streamlit** - Secondary frontend for data exploration
- **Docker Compose** - Container orchestration for all services

4.2 Product Context Diagram

```
+---------+     +----------+     +---------+
| FMP API |---->| Airflow  |---->| Postgres |
+---------+     | (DAGs)   |     | DB      |
                +----------+     +----+----+
                                      |
+---------+     +----------+     +----+----+
| DRAC    |---->| ML Model |---->| FastAPI |
| HPC     |     | (.ubj)   |     | Backend |
+---------+     +----------+     +----+----+
                                      |
                +----------+     +----+----+
                | Nginx    |<----| React   |
                | (SSL)    |     | Frontend|
                +----------+     +---------+
                                      |
                                 +---------+
                                 |Streamlit|
```

```
                                        |Frontend |
                                        +---------+
```

4.3 Assumptions and Dependencies

- FMP API remains available and within rate limits
- DRAC HPC cluster provides GPU access for model training
- DRI server remains available for production hosting
- Team has Docker and Docker Compose installed for local development
- PostgreSQL 16 is used as the sole database engine

5. Product Features

ID	Feature	Description	Priority	Status
F-01	Stock Price Prediction	XGBoost model predicts prices using 53 technical indicators	HIGH	Delivered
F-02	Confidence Scoring	Heuristic score (0-1) based on volatility, RSI, volume, return magnitude	HIGH	Delivered
F-03	Interactive Charts	Candlestick/line charts with time range selection and prediction overlay	HIGH	Delivered
F-04	Batch Prediction	"Predict All" generates forecasts for every tracked stock at once	HIGH	Delivered
F-05	FMP Data Integration	Historical OHLCV data from Financial Modeling Prep API	HIGH	Delivered
F-06	53-Feature Pipeline	RSI, MACD, Bollinger Bands, SMA/EMA, lag features, volume metrics	HIGH	Delivered
F-07	Automated Pipelines	Airflow DAGs for daily data seeding and weekly model retraining	MEDIUM	Delivered
F-08	Admin Panel	Superuser-only user management (add/edit/delete users)	MEDIUM	Delivered
F-09	User Settings	Profile editing, password change, account deletion	LOW	Delivered
F-10	Streamlit Explorer	Secondary frontend for data analysis, inference testing, datasets	MEDIUM	Delivered
F-11	DRI Deployment	Production server with SSL, Nginx, management scripts	HIGH	Delivered
F-12	Training Monitor	Real-time training job logs and progress visualization	MEDIUM	Delivered
F-13	Dark Mode	Light/dark/system theme switching	LOW	Delivered
F-14	JWT Authentication	User login/signup with JWT tokens	HIGH	Backlog
F-15	WebSocket Streaming	Real-time prediction updates via WebSocket	MEDIUM	Backlog
F-16	Prediction History	Store and track prediction accuracy over time	MEDIUM	Backlog

6. Constraints

Constraint	Description
Technology Stack	Must use Python (FastAPI), TypeScript (React), PostgreSQL as core technologies
Timeline	3 construction sprints (~10 weeks) with fixed course deadlines
Team Size	6 developers with varying availability and skill levels
Compute	GPU training limited to DRAC HPC cluster (shared resource, queue-based)
Data	FMP API has rate limits; historical data limited to available symbols
Budget	No monetary budget; all infrastructure is institution-provided
Academic	Must follow Agile/Scrum methodology with documented ceremonies

7. Quality Ranges

Quality Attribute	Target	Achieved
Performance (API p50)	< 100ms	~80ms
Performance (API p95)	< 300ms	~250ms
Inference Latency	< 50ms	~35ms
Concurrent Users	100+	Tested at 150
Backend Test Coverage	> 80%	~75%
CI Pipeline Time	< 10 min	~7 min
Model Load Time	< 2s	~1.2s
Availability	99.5%	DRI server deployed

8. Precedence and Priority

Features are prioritized using MoSCoW:

Priority	Features
Must Have	Prediction API, confidence scoring, charts, batch predict, FMP data, DRI deployment

Should Have	Airflow automation, admin panel, Streamlit explorer, training monitor
Could Have	Dark mode, user settings, dashboard optimization
Won't Have (this release)	JWT auth, WebSocket streaming, prediction history

9. Other Product Requirements

- The system must run in Docker containers for consistent deployment
- All API endpoints must be documented via Swagger UI (`/docs`)
- The CI pipeline must run on every push and PR to `main`
- Model artifacts must be versioned with `meta.json` metadata
- All database schema changes must use Alembic migrations

10. Documentation Requirements

10.1 User Manual

A comprehensive User's Manual has been prepared covering all user workflows, chart interactions, prediction interpretation, and account management. See `Documentation/Sprint4/Users_Manual.pdf`.

10.2 Online Help

- Swagger UI at `/docs` for interactive API documentation
- ReDoc at `/redoc` for formatted API reference
- In-app tooltips on chart controls, prediction cards, and confidence indicators
- Error messages with actionable descriptions

10.3 Installation Guide

A separate Installation Guide provides step-by-step instructions for Docker Compose setup, environment configuration, and production deployment. See `Documentation/Sprint5/12_Installation_Guide.pdf`.

10.4 Configuration and Read Me

- `README.md` in project root with quick start instructions
- `.env.example` with all required environment variables documented
- `development.md` with code quality tools and testing commands
- `docs_server_setup.md` for DRI production server configuration

10.5 Labelling and Packaging

- Docker images tagged with version numbers (e.g., `marketsight-backend:v1.1.0`)
- Model artifacts include `meta.json` with version, training date, feature names

- Release tags created in GitHub for each sprint milestone

Appendix: Glossary

Term	Definition
OHLCV	Open, High, Low, Close, Volume - standard market data points per trading session
XGBoost	Extreme Gradient Boosting - the ML algorithm used for price prediction
Optuna	Hyperparameter optimization framework used to tune XGBoost parameters
FMP	Financial Modeling Prep - the API providing historical market data
RSI	Relative Strength Index - momentum indicator (values 0-100)
MACD	Moving Average Convergence Divergence - trend-following indicator
Bollinger Bands	Volatility indicator using moving average with standard deviation bands
Confidence Score	Heuristic measure (0-1) of prediction reliability based on market conditions
Airflow	Apache Airflow - workflow automation for data pipelines
DAG	Directed Acyclic Graph - Airflow's pipeline definition format
DRAC	Digital Research Alliance of Canada - provides HPC compute resources
DRI	Digital Research Infrastructure - production server hosting
IOC	Initial Operational Capability - milestone where system is ready for pre-production testing
MVP	Minimum Viable Product - first functional release with core features
Ticker Symbol	Short abbreviation for a publicly traded stock (e.g., AAPL)
Sector	Broad industry classification (e.g., Technology, Healthcare)
Volatility	Measure of price fluctuation magnitude
SMA/EMA	Simple / Exponential Moving Average - trend indicators
ATR	Average True Range - volatility indicator
Lag Features	Previous period values used as prediction inputs